

— ML Performance (2026-07-01 s/d 2026-07-31)

Metric	Value
Evaluation Generated At (UTC)	2026-07-26T12:34:13.201368+00:00
=== 1. MODEL PERFORMANCE SUMMARY ===	
--- AGENUSA ---	
Active Model Version	-
Model Created At	-
Dataset Size	5000
Anomaly Count (default)	250
Anomaly Rate	5.0%
Contamination Rate	0.05
Review Score Threshold	0.004
High-Risk Score Threshold	-0.0026
Predictions at Review Boundary	499
Numeric Features Count	17
Categorical Features Count	3
Numeric Features	AMOUNT, PROCESSING_CODE, RESPONSE_CODE, MTI, TX_HOUR, TX_DAYOFWEEK, IS_NIGHT_TX, GAP_MINUTES, TERMINAL_SWITCH_FAST, AVG_AMOUNT_5, AMOUNT_OVER_AVG_RATIO, IS_DECLINED, IS_BRUTE_PATTERN, IS_MONEY_MULE_DEST, IS_HIGH_AMOUNT_PATTERN, MIDNIGHT_AMOUNT_SPIKE, RAPID_RETRY_DECLINED
Categorical Features	TERMINAL_ID, MERCHANT_ID, DEST_ACCOUNT_NUMBER

--- NUSABILL ---	
Active Model Version	-
Model Created At	-
Dataset Size	5000
Anomaly Count (default)	250
Anomaly Rate	5.0%
Contamination Rate	0.05
Review Score Threshold	0.0357
High-Risk Score Threshold	-0.0298
Predictions at Review Boundary	501
Numeric Features Count	12
Categorical Features Count	2
Numeric Features	BILL_AMOUNT, PAYMENT_AMOUNT, REFUND_FLAG, PAYMENT_DELAY_DAYS, PAYMENT_TO_BILL_RATIO, UNDERPAY_FLAG, HIGH_SPIKE_FLAG, CHANNEL_API_FLAG, PAYMENT_GAP_MINUTES, BURST_FLAG, CHANNEL_SWITCH_TO_API, EARLY_PAYMENT_ANOMALY
Categorical Features	CHANNEL, BILL_STATUS
=== 2. RETRAIN HISTORY ===	Periode: 2026-07-01 s/d 2026-07-31

Metric	Value
Total Retrain Executions	0
Successful Retrains	0
Retrain Success Rate (%)	0
Total Records Trained (cumulative)	0
Total Feedback Records Used (cumulative)	0
Total Anomalies Found (cumulative)	0
Total New Patterns Discovered (cumulative)	0
Trigger Type Breakdown	
(tidak ada retrain pada periode ini)	
Recent Retrain Executions (max 10)	
Execution Time Trigger Status Records Feedback Used New Patterns	
=== 3. PATTERN DISCOVERY ===	
Top 10 Patterns Category Risk Score Hit Count	
Agenusa - EDC Terminal Pooling & Card Washing Detection	category=Merchant Collusion & Terminal Abuse, risk_score=90, hits=32
Agenusa - Account Takeover & Brute Force PIN Guessing	category=Account Takeover Suspect, risk_score=75, hits=14
Agenusa - Velocity Burst (High-Frequency Card Usage)	category=Velocity Attack & Shopping Spree, risk_score=85, hits=14
Agenusa - Critical Card Testing Burst Detection	category=Carding & Brute Force, risk_score=95, hits=10
Agenusa - Fan-In Syndicate (Massive Card Pooling)	category=Merchant Collusion & Syndicate, risk_score=100, hits=4
Nusabill - Tuned Fake Invoice Blast Detection (Low Nominal Trap)	category=Billing Scam & Bulk Invoicing, risk_score=85, hits=3
Nusabill - Fan-Out Spam (Fake Invoice Mass Blast)	category=Phishing & Spam, risk_score=80, hits=3
Nusabill - High-Velocity Split Payment Anomaly (Smurfing)	category=Money Laundering & Split Transaction, risk_score=80, hits=2
Agenusa - Super Pattern (Advanced Syndicate Attack)	category=Syndicate & Multi-Vector Attack, risk_score=100, hits=2
Nusabill - Velocity Burst (API Abuse / Bulk Anomaly)	category=System Abuse & Rate Limit Evasion, risk_score=75, hits=0
Total Patterns in System	10
Active Patterns	10
=== 4. FRAUD DETECTION EFFECTIVENESS ===	Periode: 2026-07-01 s/d 2026-07-31
--- AGENUSA ---	
Total Transactions Processed	135
Actual Fraud Count	52
Actual Fraud Rate (%)	38.52
Model Anomaly Rate (%)	5.0
Anomaly Rate vs Actual Fraud Rate Gap	-33.52 pp (model kurang sensitif)
--- NUSABILL ---	
Total Transactions Processed	56
Actual Fraud Count	4

Metric	Value
Actual Fraud Rate (%)	7.14
Model Anomaly Rate (%)	5.0
Anomaly Rate vs Actual Fraud Rate Gap	-2.14 pp (model kurang sensitif)
Total New Patterns Generated (all retrains, periode ini)	0
=== CATATAN ===	
Akurasi Model	Tidak tersedia — dataset training bersifat unlabeled (tanpa kolom IS_FRAUD), sehingga metrik Precision, Recall, dan F1-Score tidak dapat dihitung.
Metode Evaluasi	Berbasis threshold boundary dari hasil training (Isolation Forest), bukan validasi terhadap ground truth.
Anomaly Rate vs Fraud Rate	Perbandingan ini bersifat indikatif — anomaly rate adalah parameter model (contamination), sedangkan fraud rate adalah hasil keputusan final (rule engine + pattern + ML + review analisis).